

1. What 3 things causes signal loss in wires?

- Material of wire
- Length of wire
- Skin effect

2. When do you start needing transmission lines?

When the distance between the transmission and receiver is too big

3. What is a transmission line?

wires made to transmit data efficiently based on requirements

4. A 50Ω (Z_0) line is connected to an open circuit what is the reflection coefficient Γ

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \text{ and open circuit has } \infty \text{ resistance}$$

$$\Gamma = \frac{\infty - 50}{\infty + 50} = 1$$

5. A 50Ω (Z_0) line is connected to a short circuit what is the reflection coefficient Γ

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \text{ and short circuit has } 0 \text{ resistance}$$

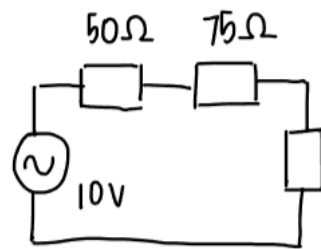
$$\Gamma = \frac{0 - 50}{0 + 50} = -1$$

6. A 50Ω line is connected to a matched load (50Ω), what is Γ ?

$$Z_0 = Z_L = 50\Omega$$

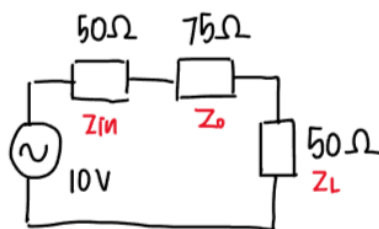
$$\Gamma = \frac{50 - 50}{50 + 50} = 0$$

7. A signal generator with output $R=50\Omega$ is connected to a 75Ω line. How much voltage is delivered onto the line if its output is $10V$?



$$V_{Z_0} = \frac{75}{50+75} (10) = 6V$$

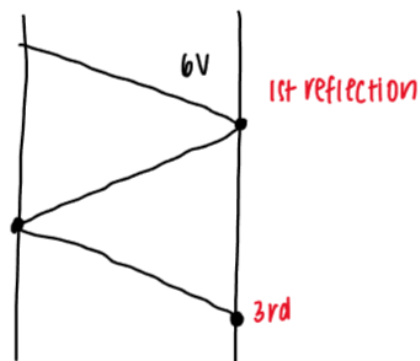
8. If the line in previous question is connected to a 50Ω load. Calculate the size of the first 3 reflections on the line.



$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = -0.2$$

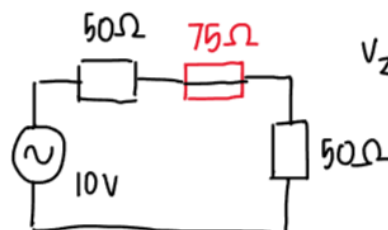
$$\Gamma_S = \frac{Z_{in} - Z_0}{Z_{in} + Z_0} = -0.2$$

2nd



$$\begin{aligned} \text{1st reflection} &= 6(-0.2) = -1.2V \\ \text{2nd reflection} &= 6(-0.2)^2 = 0.24V \\ \text{3rd reflection} &= 6(-0.2)^3 = -0.048V \end{aligned}$$

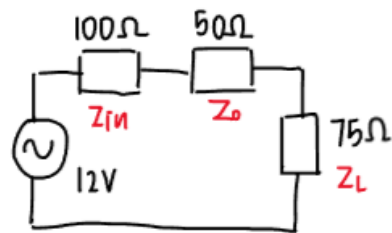
9. What voltage will the line settle to?



$$V_{Z_L} = \frac{50}{50+50} (10) = 5V$$

Line will settle to 5V

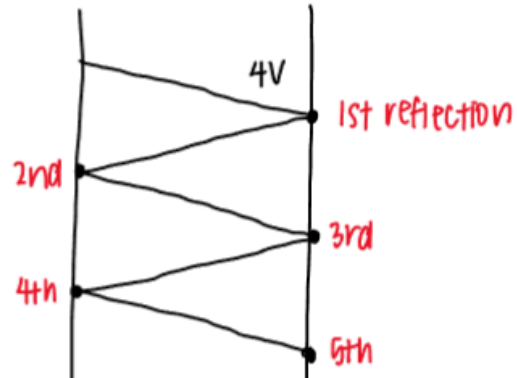
10. A signal generator with output $R=100\Omega$ is connected to a 50Ω line with a 75Ω load connected and voltage of $12V$. Calculate the first 5 reflections and find the settled voltage of the line.



$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = 0.2$$

$$\Gamma_S = \frac{Z_{in} - Z_0}{Z_{in} + Z_0} = \frac{1}{3} = 0.33$$

$$V_{Z_0} = \frac{50}{100+50} (12) = 4V$$



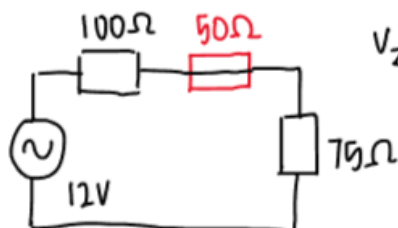
$$1st \text{ reflection} = 4(0.2) = 0.8$$

$$2nd \text{ reflection} = 4(0.2)(1/3) = 0.27$$

$$3rd \text{ reflection} = 4(0.2)^2(1/3) = 0.053$$

$$4th \text{ reflection} = 4(0.2)^2(1/3)^2 = 0.018$$

$$5th \text{ reflection} = 4(0.2)^3(1/3)^2 = 0.0036$$



$$V_{Z_L} = \frac{75}{100+75} (12) = 5.14V$$

Line will settle to 5.14V